

Cuaderno de Prácticas

Álgebra

Grado en Ingeniería Informática

CURSO 2024/25

Datos personales

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Grupo de teoría : A

Grupo de prácticas :

Práctica 9 sesión 8

Código

```
In[*]:=
(**)
el1[i_,j_,n_]:=Module[{B},
    B=IdentityMatrix[n];
    B[[i,i]]=0;
    B[[j,j]]=0;
    B[[i,j]]=1;
    B[[j,i]]=B[[i,j]];
    B];

(**)
el2[i_,s_,n_]:=Module[{B},
    B=IdentityMatrix[n];
    B[[i,i]]=s;
    B];

(**)
el3[i_,j_,t_,n_]:=Module[{B},
    B=IdentityMatrix[n];
    B[[i,j]]=t;
    B];

In[*]:=
HERMITE[matriz_,opciones_]:=Module[{pivote,aux,col,i,j,t,kt,Adib,A,transformaciones,n,m,el1,e2},
    A=matriz;m=Dimensions[A][[1]];n=Dimensions[A][[2]];
    (*Definimos las matrices elementales*)
```

```

el1[i_,j_] := Module[{B}, B = IdentityMatrix[m]; B[[i,i]] = 0; B[[j,j]] = 0; B[[i,j]] = 1; B[[j,i]] = B[[i,j]]; B];
el2[i_,s_] := Module[{B}, B = IdentityMatrix[m]; B[[i,i]] = s; B];
el3[i_,j_,t_] := Module[{B}, B = IdentityMatrix[m]; B[[i,j]] = t; B];
transformaciones = {};
A = matriz;
If[opciones == 1,
Print[Style["CÁLCULO DE LA FORMA NORMAL DE HERMITE POR FILAS DE ", {Bold}], A // MatrixForm];
Print["1. Calculamos la matriz escalonada:"];
];
pivote = 1; t = 1; col = 0;
While[pivote ≠ 0,
If[t < m && (t + col) ≤ n,
If[opciones == 1,
Print[" Buscamos el pivote ", t, "º:"]; (*Buscamos siguiente pivote*)
];
j = t + col; i = t;
While[A[[i,j]] == 0, If[i < m, i++;, If[j < m, i = t; j++; col++;, Break[]];];
pivote = A[[i,j]];
If[A[[i,j]] ≠ 0,
If[t ≠ i && i ≤ m,
aux = A[[t]]; A[[t]] = A[[i]]; A[[i]] = aux; (*Intercambio de filas para subir el pivote*)
AppendTo[transformaciones, el1[i, t]];
If[opciones == 1,
Print[" Lo ponemos en la fila ", t, "ª"];
Adib = A;
Do[Adib[[i,j]] = Style[Adib[[i,j]], {Italic, Bold, Blue}];, {j, 1, n}];
Do[Adib[[t,j]] = Style[Adib[[t,j]], {Bold, Blue}];, {j, 1, n}];
Print[" * ", Style["Transf. elemental tipo I", {Blue}], ", cambiar", " fila ", Style[i, {Italic, Bold}], " por la fila ", Style[t, {Italic, Bold}], ""];
Adib // MatrixForm];
];
];
If[A[[t,j]] ≠ 1,
kt = (1/A[[t,j]]); A[[t]] = kt * A[[t]]; (*Multiplificamos la fila por el inverso del pivote *)
AppendTo[transformaciones, el2[t, kt]];
If[opciones == 1,
Adib = A;
Do[Adib[[t,j]] = Style[Adib[[t,j]], {Bold, Red}];, {j, 1, n}];
Print[" Hacemos 1 el pivote."];
Print[" * ", Style["Transf. elemental tipo II", {Red}], ", la fila ", Style[t, {Bold, Red}], " por la fila ", Style[t, {Bold, Red}], ""];
];
];
];
(*Mostramos el pivote*)
If[opciones == 1,
Adib = A;
Adib[[t,j]] = Style[Adib[[t,j]], {Bold, Green}];
Print[" ---> Pivote ", t, "º: ", Adib // MatrixForm];
(*Hacemos ceros por debajo del pivote*)
Print[" - Hacemos ceros por debajo del pivote ", t, "º:"];
];
Do[
If[A[[k,j]] ≠ 0,
kt = (-A[[k,j]]);
A[[k]] = A[[k]] + (-A[[k,j]] * A[[t]]);
];
];

```

```

AppendTo[transformaciones,el3[k,t,kt]];
If[opciones==1,
Adib=A;
Do[Adib[[k,j]]=Style[Adib[[k,j]],{Italic,Bold,Orange}];,{j,1,n}];
Do[Adib[[t,j]]=Style[Adib[[t,j]],{Bold,Orange}];,{j,1,n}];
Print[" * ",Style["T. elem. tipo III",{Orange}], " a la fila ",Style[k,{Italic,Bold,Orange}
Adib//MatrixForm];
];(*Hacemos ceros por debajo del pivote*)
];
,{k,t+1,m}];
,col=0;
If[t<n,
While[A[[t,t+col]]==0 ,col++;If[(t+col)==n,Break[]];];];
If[(j+col)<=n && A[[t,t+col]]!=0,
kt=(1/A[[t,t+col]]);
A[[t]]=(1/A[[t,t+col]])*A[[t]];
AppendTo[transformaciones,el2[t,kt]];
If[opciones==1,
Print[" Buscamos el pivote ",t,"º:"];
Print[" Hacemos 1 el pivote."];
Adib=A;
Do[Adib[[t,j]]=Style[Adib[[t,j]],{Bold,Red}];,{j,1,n}];
Print[" * ",Style["Transf. elemental tipo II",{Red}], " la fila ",Style[t,{Bold,Red}], " p
Adib//MatrixForm];
];
(*Hacemos 1 el pivote*);];
pivote=0;
];
t++;
If[t>n || t>m,pivote=0];];
];
If[opciones==1,
Print["2. Calculamos la matriz escalonada reducida, hacemos cero por encima de los pivotes:"];
];
(*Ya tenemos la matriz escalonada, ahora hacemos ceros por debajo de los pivotes*)
i=2;j=2;pivote=1;
While[pivote!=0,
If[opciones==1,
Print[" Pivote ",i,"º:"];
];
While[A[[i,j]]!=1 && j<=n,j++;If[j>n,Break[]];];(*Buscamos el pivote en la fila*)×
If[j<=n,
Do[
If[A[[k,j]]!=0,
kt=(-A[[k,j]]);A[[k]]=A[[k]]+(-A[[k,j]])* A[[i]]; (*A[[k,j]]=0*)
AppendTo[transformaciones,el3[k,i,kt]];
If[opciones==1,
Adib=A;
Do[Adib[[k,j]]=Style[Adib[[k,j]],{Italic,Bold,Orange}];,{j,1,n}];
Do[Adib[[i,j]]=Style[Adib[[i,j]],{Bold,Orange}];,{j,1,n}];

Print[" * ",Style["T. elem. tipo III",{Orange}], " a la fila ",Style[k,{Italic,Bold,Orange}
Adib//MatrixForm];
];(*Hacemos ceros por encima del pivote*)
];
];

```

```

, {k, 1, i-1}];
i++;
j++;
If[i > m || j > n, pivote = 0;];
, pivote = 0;];];
If[opciones == 1,
Print["FORMA NORMAL DE HERMITE: ", A // MatrixForm];
];
If[opciones != 2, Return[A],
Q = transformaciones[[1]];
Do[Q = transformaciones[[i]].Q
, {i, 2, Length[transformaciones]}];
Print["Matriz regular Q tal que Q.A = H, "];
Print["Q = ", Q // MatrixForm];
Print["Lista de matrice elementales: ", MatrixForm /@ transformaciones];
Return[transformaciones]];
];

```

Ejercicio 8: Rel 3

Dadas las matrices

$$\begin{aligned}
 \text{In[*]} := \quad m8A &= \begin{pmatrix} 1 & 2 & 1 & 0 \\ 0 & 1 & 1 & -1 \\ 1 & 2 & -1 & 2 \\ 1 & 1 & 0 & 1 \end{pmatrix}; \\
 m8H &= \begin{pmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & -1 \\ 0 & 0 & 0 & 0 \end{pmatrix} \\
 m8C &= \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 1 & -1 & 0 & 0 \end{pmatrix}
 \end{aligned}$$

Out[*] = { {1, 0, 0, 1}, {0, 1, 0, 0}, {0, 0, 1, -1}, {0, 0, 0, 0} }

Out[*] = { {1, 0, 0, 0}, {0, 1, 0, 0}, {0, 0, 1, 0}, {1, -1, 0, 0} }

a) Calcular una matriz regular Q tal que QA=H

```
In[*]:= A=  $\begin{pmatrix} 1 & 2 & 1 & 0 \\ 0 & 1 & 1 & -1 \\ 1 & 2 & -1 & 2 \\ 1 & 1 & 0 & 1 \end{pmatrix}$ ;
H=RowReduce[A];
H//MatrixForm
AI=  $\begin{pmatrix} 1 & 2 & 1 & 0 & 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & -1 & 0 & 1 & 0 & 0 \\ 1 & 2 & -1 & 2 & 0 & 0 & 1 & 0 \\ 1 & 1 & 0 & 1 & 0 & 0 & 0 & 1 \end{pmatrix}$ ;
A1=e13[3,1,-1,4].AI;
A1//MatrixForm
```

```
Out[*]//MatrixForm=
 $\begin{pmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & -1 \\ 0 & 0 & 0 & 0 \end{pmatrix}$ 
```

```
Out[*]//MatrixForm=
 $\begin{pmatrix} 1 & 2 & 1 & 0 & 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & -1 & 0 & 1 & 0 & 0 \\ 0 & 0 & -2 & 2 & -1 & 0 & 1 & 0 \\ 1 & 1 & 0 & 1 & 0 & 0 & 0 & 1 \end{pmatrix}$ 
```

```
In[*]:= A2=e13[4,1,-1,4].A1;
A2//MatrixForm
```

```
Out[*]//MatrixForm=
 $\begin{pmatrix} 1 & 2 & 1 & 0 & 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & -1 & 0 & 1 & 0 & 0 \\ 0 & 0 & -2 & 2 & -1 & 0 & 1 & 0 \\ 0 & -1 & -1 & 1 & -1 & 0 & 0 & 1 \end{pmatrix}$ 
```

```
In[*]:= A3=e13[4,2,1,4].A2;
A3//MatrixForm
```

```
Out[*]//MatrixForm=
 $\begin{pmatrix} 1 & 2 & 1 & 0 & 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & -1 & 0 & 1 & 0 & 0 \\ 0 & 0 & -2 & 2 & -1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & -1 & 1 & 0 & 1 \end{pmatrix}$ 
```

```
In[*]:= A4 = e12[3, -1/2, 4].A3;
A4 // MatrixForm
[forma de matriz]
```

```
Out[*]//MatrixForm=
 $\begin{pmatrix} 1 & 2 & 1 & 0 & 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & -1 & 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & -1 & \frac{1}{2} & 0 & -\frac{1}{2} & 0 \\ 0 & 0 & 0 & 0 & -1 & 1 & 0 & 1 \end{pmatrix}$ 
```

```
In[ ]:= A5 = e13[2, 3, -1, 4].A4;
A5 // MatrixForm
|forma de matriz
```

```
Out[ ]//MatrixForm=

$$\begin{pmatrix} 1 & 2 & 1 & 0 & 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & -\frac{1}{2} & 1 & \frac{1}{2} & 0 \\ 0 & 0 & 1 & -1 & \frac{1}{2} & 0 & -\frac{1}{2} & 0 \\ 0 & 0 & 0 & 0 & -1 & 1 & 0 & 1 \end{pmatrix}$$

```

```
In[ ]:= A6=e13[1,3,-1,4].A5;
A6//MatrixForm
```

```
Out[ ]//MatrixForm=

$$\begin{pmatrix} 1 & 2 & 0 & 1 & \frac{1}{2} & 0 & \frac{1}{2} & 0 \\ 0 & 1 & 0 & 0 & -\frac{1}{2} & 1 & \frac{1}{2} & 0 \\ 0 & 0 & 1 & -1 & \frac{1}{2} & 0 & -\frac{1}{2} & 0 \\ 0 & 0 & 0 & 0 & -1 & 1 & 0 & 1 \end{pmatrix}$$

```

```
In[ ]:= A7 = e13[1, 2, -2, 4].A6;
A7 // MatrixForm
|forma de matriz
```

```
Out[ ]//MatrixForm=

$$\begin{pmatrix} 1 & 0 & 0 & 1 & \frac{3}{2} & -2 & -\frac{1}{2} & 0 \\ 0 & 1 & 0 & 0 & -\frac{1}{2} & 1 & \frac{1}{2} & 0 \\ 0 & 0 & 1 & -1 & \frac{1}{2} & 0 & -\frac{1}{2} & 0 \\ 0 & 0 & 0 & 0 & -1 & 1 & 0 & 1 \end{pmatrix}$$

```

```
In[ ]:= Q = 
$$\begin{pmatrix} \frac{3}{2} & -2 & -\frac{1}{2} & 0 \\ -\frac{1}{2} & 1 & \frac{1}{2} & 0 \\ \frac{1}{2} & 0 & -\frac{1}{2} & 0 \\ -1 & 1 & 0 & 1 \end{pmatrix};$$

Det[Q]
|determinante
Inverse[Q] // MatrixForm
|matriz inversa |forma de matriz
H == Q.A
```

```
Out[ ]:=

$$-\frac{1}{2}$$

```

```
Out[ ]//MatrixForm=

$$\begin{pmatrix} 1 & 2 & 1 & 0 \\ 0 & 1 & 1 & 0 \\ 1 & 2 & -1 & 0 \\ 1 & 1 & 0 & 1 \end{pmatrix}$$

```

```
Out[ ]:=
True
```

```
In[*]:= HERMITE[A,2];
```

Matriz regular Q tal que $Q.A = H$,

$$Q = \begin{pmatrix} \frac{3}{2} & -2 & -\frac{1}{2} & 0 \\ -\frac{1}{2} & 1 & \frac{1}{2} & 0 \\ \frac{1}{2} & 0 & -\frac{1}{2} & 0 \\ -1 & 1 & 0 & 1 \end{pmatrix}$$

Lista de matrice elementales: $\left\{ \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ -1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}, \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ -1 & 0 & 0 & 1 \end{pmatrix}, \right.$

$$\left. \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \end{pmatrix}, \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & -\frac{1}{2} & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}, \begin{pmatrix} 1 & -2 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}, \begin{pmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}, \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \right\}$$

b) Calcular una matriz regular P tal que $AP=C$

```
In[*]:= BI=
  {1 0 1 1 1 0 0 0}
  {2 1 2 1 0 1 0 0};
  {1 1 -1 0 0 0 1 0}
  {0 -1 2 1 0 0 0 1}
```

```
TB1=e13[2,1,-2,4];
```

```
TB2=e13[3,1,-1,4];
```

```
B1=TB1.TB2.BI;
```

```
B1//MatrixForm
```

```
Out[*]//MatrixForm=
```

$$\begin{pmatrix} 1 & 0 & 1 & 1 & 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & -1 & -2 & 1 & 0 & 0 \\ 0 & 1 & -2 & -1 & -1 & 0 & 1 & 0 \\ 0 & -1 & 2 & 1 & 0 & 0 & 0 & 1 \end{pmatrix}$$

```
In[*]:= TB3=e13[3,2,-1,4];
```

```
TB4=e13[4,2,1,4];
```

```
B2=TB3.TB4.B1;
```

```
B2//MatrixForm
```

```
Out[*]//MatrixForm=
```

$$\begin{pmatrix} 1 & 0 & 1 & 1 & 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & -1 & -2 & 1 & 0 & 0 \\ 0 & 0 & -2 & 0 & 1 & -1 & 1 & 0 \\ 0 & 0 & 2 & 0 & -2 & 1 & 0 & 1 \end{pmatrix}$$

```
In[*]:= TB5=e12[3,-1/2,4];
B3=TB5.B2;
B3//MatrixForm
```

```
Out[*]//MatrixForm=
```

$$\begin{pmatrix} 1 & 0 & 1 & 1 & 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & -1 & -2 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 & -\frac{1}{2} & \frac{1}{2} & -\frac{1}{2} & 0 \\ 0 & 0 & 2 & 0 & -2 & 1 & 0 & 1 \end{pmatrix}$$

```
In[*]:= TB6=e13[1,3,-1,4];
TB7=e13[4,3,-2,4];
B4=TB7.TB6.B3;
B4//MatrixForm
```

```
Out[*]//MatrixForm=
```

$$\begin{pmatrix} 1 & 0 & 0 & 1 & \frac{3}{2} & -\frac{1}{2} & \frac{1}{2} & 0 \\ 0 & 1 & 0 & -1 & -2 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 & -\frac{1}{2} & \frac{1}{2} & -\frac{1}{2} & 0 \\ 0 & 0 & 0 & 0 & -1 & 0 & 1 & 1 \end{pmatrix}$$

```
In[*]:= Ab=
```

$$\begin{pmatrix} 1 & 0 & 1 & 1 \\ 2 & 1 & 2 & 1 \\ 1 & 1 & -1 & 0 \\ 0 & -1 & 2 & 1 \end{pmatrix};$$

```
Cb=
```

$$\begin{pmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix};$$

```
Pb=
```

$$\begin{pmatrix} \frac{3}{2} & -\frac{1}{2} & \frac{1}{2} & 0 \\ -2 & 1 & 0 & 0 \\ -\frac{1}{2} & \frac{1}{2} & -\frac{1}{2} & 0 \\ -1 & 0 & 1 & 1 \end{pmatrix};$$

```
Ab.Pb==Cb
```

```
Out[*]=
```

False

```
In[*]:= HERMITE[Ab,2]
```


Matriz regular Q tal que $Q.A = H$,

$$Q = \begin{pmatrix} \frac{3}{2} & -\frac{1}{2} & \frac{1}{2} & 0 \\ -2 & 1 & 0 & 0 \\ -\frac{1}{2} & \frac{1}{2} & -\frac{1}{2} & 0 \\ -1 & 0 & 1 & 1 \end{pmatrix}$$

Lista de matrice elementales: $\left\{ \begin{pmatrix} 1 & 0 & 0 & 0 \\ -2 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}, \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ -1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}, \right.$

$$\left. \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & -1 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}, \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \end{pmatrix}, \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & -\frac{1}{2} & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}, \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & -2 & 1 \end{pmatrix}, \begin{pmatrix} 1 & 0 & -1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \right\}$$

Out[8]=

```
{ {{1, 0, 0, 0}, {-2, 1, 0, 0}, {0, 0, 1, 0}, {0, 0, 0, 1}},
  {{1, 0, 0, 0}, {0, 1, 0, 0}, {-1, 0, 1, 0}, {0, 0, 0, 1}},
  {{1, 0, 0, 0}, {0, 1, 0, 0}, {0, -1, 1, 0}, {0, 0, 0, 1}},
  {{1, 0, 0, 0}, {0, 1, 0, 0}, {0, 0, 1, 0}, {0, 1, 0, 1}},
  {{1, 0, 0, 0}, {0, 1, 0, 0}, {0, 0, -1/2, 0}, {0, 0, 0, 1}},
  {{1, 0, 0, 0}, {0, 1, 0, 0}, {0, 0, 1, 0}, {0, 0, -2, 1}},
  {{1, 0, -1, 0}, {0, 1, 0, 0}, {0, 0, 1, 0}, {0, 0, 0, 1}} }
```

c)¿Son H y C equivalentes? ¿Por qué?

Ejercicio 10 : Rel 3

Calcular el rango de las matrices

a)

DNI: ...8082

a12=2

In[9]:=

$$m10a = \begin{pmatrix} 1 & 2 & 1 & 0 \\ 2 & 4 & -2 & 8 \\ -1 & 0 & 1 & -2 \end{pmatrix}$$

Out[9]=

```
{{1, 2, 1, 0}, {2, 4, -2, 8}, {-1, 0, 1, -2}}
```

In[10]:=

MatrixRank[m10a]

Out[10]=

3

b)

```
In[ ]:= m10b=
```

$$\begin{pmatrix} 1 & -1 & 1 \\ 2 & 1 & 7 \\ 0 & 1 & 2 \\ 1 & 0 & 3 \end{pmatrix}$$

```
Out[ ]:=
```

$$\{\{1, -1, 1\}, \{2, 1, 7\}, \{0, 1, 2\}, \{1, 0, 3\}\}$$

```
In[ ]:= MatrixRank[m10b]
```

```
Out[ ]:=
```

3

c)

```
In[ ]:= m10c=
```

$$\begin{pmatrix} 5 & 0 & 20 & -10 \\ 1 & 0 & 4 & -2 \\ 2 & 0 & 8 & -4 \end{pmatrix}$$

```
Out[ ]:=
```

$$\{\{5, 0, 20, -10\}, \{1, 0, 4, -2\}, \{2, 0, 8, -4\}\}$$

```
In[ ]:= MatrixRank[m10c]
```

```
Out[ ]:=
```

1

d)

```
In[ ]:= m10d=
```

$$\begin{pmatrix} 3 & 6 & -5 & 0 \\ 1 & 2 & 2 & 9 \\ 2 & 4 & -3 & 1 \end{pmatrix}$$

```
Out[ ]:=
```

$$\{\{3, 6, -5, 0\}, \{1, 2, 2, 9\}, \{2, 4, -3, 1\}\}$$

```
In[ ]:= MatrixRank[m10d]
```

```
Out[ ]:=
```

3

Ejercicio 12 : Rel 3

Hallar las inversas de las matrices:

a)

```
In[*]:= m12a=
```

$$\begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 3 & 5 & -2 \\ 3 & 5 & 1 & 1 \\ 1 & 1 & -5 & -7 \end{pmatrix}$$

```
Out[*]=
```

$$\{\{1, 2, 3, 4\}, \{2, 3, 5, -2\}, \{3, 5, 1, 1\}, \{1, 1, -5, -7\}\}$$

```
In[*]:= Det[m12a]
```

```
Out[*]=
```

$$0$$

Como el determinante es 0 no tiene inversa

b)

```
In[1]:= m12b=
```

$$\begin{pmatrix} 1 & 2 & 3 & 1 \\ 1 & 3 & 3 & 2 \\ 2 & 4 & 3 & 3 \\ 1 & 1 & 1 & 1 \end{pmatrix}$$

```
Out[1]=
```

$$\{\{1, 2, 3, 1\}, \{1, 3, 3, 2\}, \{2, 4, 3, 3\}, \{1, 1, 1, 1\}\}$$

```
In[*]:= Det[m12b]
```

```
Out[*]=
```

$$-1$$

```
In[*]:= Inverse[m12b]
```

```
Out[*]=
```

$$\{\{1, -2, 1, 0\}, \{1, -2, 2, -3\}, \{0, 1, -1, 1\}, \{-2, 3, -2, 3\}\}$$

```
In[*]:= {{1,-2,1,0},{1,-2,2,-3},{0,1,-1,1},{-2,3,-2,3}}//MatrixForm
```

```
Out[*]//MatrixForm=
```

$$\begin{pmatrix} 1 & -2 & 1 & 0 \\ 1 & -2 & 2 & -3 \\ 0 & 1 & -1 & 1 \\ -2 & 3 & -2 & 3 \end{pmatrix}$$

```
In[8]:= n12b=Dimensions[m12b][[1];
S12b=Minors[m12b,n12b-1];
S12b//MatrixForm
```

```
Out[10]//MatrixForm=
```

$$\begin{pmatrix} -3 & 1 & 3 & 0 \\ -2 & 1 & 2 & -1 \\ -3 & 1 & 2 & -2 \\ -2 & 0 & 1 & -1 \end{pmatrix}$$

```
In[11]:= adj12b=Table[ ( (-1)^(i+j) *S12b[[n12b-i+1,n12b-j+1]] ) [[1]],{i,1,4},{j,1,4}];
MatrixForm[adj12b]
```

```
Out[12]//MatrixForm=

$$\begin{pmatrix} (-1) \llbracket 1 \rrbracket & (-1) \llbracket 1 \rrbracket & 0 \llbracket 1 \rrbracket & 2 \llbracket 1 \rrbracket \\ 2 \llbracket 1 \rrbracket & 2 \llbracket 1 \rrbracket & (-1) \llbracket 1 \rrbracket & (-3) \llbracket 1 \rrbracket \\ (-1) \llbracket 1 \rrbracket & (-2) \llbracket 1 \rrbracket & 1 \llbracket 1 \rrbracket & 2 \llbracket 1 \rrbracket \\ 0 \llbracket 1 \rrbracket & 3 \llbracket 1 \rrbracket & (-1) \llbracket 1 \rrbracket & (-3) \llbracket 1 \rrbracket \end{pmatrix}$$

```

```
In[13]:= inv12b= 
$$\frac{\text{Transpose}[\text{adj12b}]}{\text{Det}[\text{m12b}]};$$

MatrixForm[inv12b]
```

```
Out[14]//MatrixForm=

$$\begin{pmatrix} -(-1) \llbracket 1 \rrbracket & -2 \llbracket 1 \rrbracket & -(-1) \llbracket 1 \rrbracket & -0 \llbracket 1 \rrbracket \\ -(-1) \llbracket 1 \rrbracket & -2 \llbracket 1 \rrbracket & -(-2) \llbracket 1 \rrbracket & -3 \llbracket 1 \rrbracket \\ -0 \llbracket 1 \rrbracket & -(-1) \llbracket 1 \rrbracket & -1 \llbracket 1 \rrbracket & -(-1) \llbracket 1 \rrbracket \\ -2 \llbracket 1 \rrbracket & -(-3) \llbracket 1 \rrbracket & -2 \llbracket 1 \rrbracket & -(-3) \llbracket 1 \rrbracket \end{pmatrix}$$

```

c)

```
In[15]:= m12c= 
$$\begin{pmatrix} 1 & 2-i & -1+2i \\ 1 & 2 & -2+i \\ -1 & -2+i & 2-2i \end{pmatrix}$$

```

```
Out[15]=

$$\{\{1, 2-i, -1+2i\}, \{1, 2, -2+i\}, \{-1, -2+i, 2-2i\}\}$$

```

```
In[*]:= Det[m12c]
```

```
Out[*]=
i
```

Si i = 0 no tiene inversa y si i es != 0 tiene inversa

```
In[*]:= Inverse[m12c]
```

```
Out[*]=

$$\left\{ \left\{ -i, \frac{-2+i}{i}, \frac{-2-i^2}{i} \right\}, \left\{ 1, \frac{1}{i}, \frac{1+i}{i} \right\}, \{1, 0, 1\} \right\}$$

```

```
In[*]:= 
$$\left\{ \left\{ -i, \frac{-2+i}{i}, \frac{-2-i^2}{i} \right\}, \left\{ 1, \frac{1}{i}, \frac{1+i}{i} \right\}, \{1, 0, 1\} \right\} // \text{MatrixForm}$$

```

```
Out[*]//MatrixForm=

$$\begin{pmatrix} -i & \frac{-2+i}{i} & \frac{-2-i^2}{i} \\ 1 & \frac{1}{i} & \frac{1+i}{i} \\ 1 & 0 & 1 \end{pmatrix}$$

```

```
In[16]:= n12c=Dimensions[m12c][[1]];
S12c=Minors[m12c,n12c-1];
S12c//MatrixForm
```

Out[18]//MatrixForm=

$$\begin{pmatrix} i & -1-i & -2-i^2 \\ 0 & 1 & 2-i \\ i & -i & -i^2 \end{pmatrix}$$

```
In[19]:= adj12c=Table[(-1)^(i+j)*S12c[[n12c-i+1,n12c-j+1]][[1]],{i,1,4},{j,1,4}];
MatrixForm[adj12c]
```

Part: Part specification $(-1)[[1]]$ is longer than depth of object. [i](#)

Part: Part specification $1[[1]]$ is longer than depth of object. [i](#)

Part: Part specification $1[[1]]$ is longer than depth of object. [i](#)

General: Further output of Part::partd will be suppressed during this calculation. [i](#)

Out[20]//MatrixForm=

$$\begin{pmatrix} (-1)[[1]] & 1[[1]] & 1[[1]] & -1 \\ 0[[1]] & 1[[1]] & 0[[1]] & \text{List}[[1]] \\ (-11)[[1]] & 4[[1]] & 3[[1]] & -1 \\ -1 & \{4, -5, -18\}, \{0, 1, -2\}, \{4, -4, -16\} & -1 & \text{Symbol}[[1]] \end{pmatrix}$$

```
In[23]:= inv12c=Transpose[adj12c];
Det[m12c];
MatrixForm[inv12c]
```

Out[24]//MatrixForm=

$$\begin{pmatrix} \frac{(-1)[[1]]}{i} & \frac{0[[1]]}{i} & \frac{(-11)[[1]]}{i} & -\frac{1}{i} \\ \frac{1[[1]]}{i} & \frac{1[[1]]}{i} & \frac{4[[1]]}{i} & \left\{ \left\{ \frac{4}{i}, -\frac{5}{i}, -\frac{18}{i} \right\}, \left\{ 0, \frac{1}{i}, -\frac{2}{i} \right\}, \left\{ \frac{4}{i}, -\frac{4}{i}, -\frac{16}{i} \right\} \right\} \\ \frac{1[[1]]}{i} & \frac{0[[1]]}{i} & \frac{3[[1]]}{i} & -\frac{1}{i} \\ -\frac{1}{i} & \frac{\text{List}[[1]]}{i} & -\frac{1}{i} & \frac{\text{Symbol}[[1]]}{i} \end{pmatrix}$$

d)

```
In[25]:= m12d=

$$\begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 4 \\ 3 & 4 & 6 \end{pmatrix}$$

```

Out[25]=

$\{\{1, 2, 3\}, \{2, 3, 4\}, \{3, 4, 6\}\}$

```
In[26]:= Det[m12d]
```

Out[26]=

-1

```
In[*]:= Dimensions[m12d]
```

```
Out[*]=  
{3, 3}
```

```
In[*]:= Inverse[m12d]
```

```
Out[*]=  
{{-2, 0, 1}, {0, 3, -2}, {1, -2, 1}}
```

```
In[*]:= {{-2,0,1},{0,3,-2},{1,-2,1}}//MatrixForm
```

```
Out[*]//MatrixForm=  

$$\begin{pmatrix} -2 & 0 & 1 \\ 0 & 3 & -2 \\ 1 & -2 & 1 \end{pmatrix}$$

```

```
In[26]:= n12d=Dimensions[m12d][[1];  
S12d=Minors[m12d,n12d-1];  
S12d//MatrixForm
```

```
Out[28]//MatrixForm=  

$$\begin{pmatrix} -1 & -2 & -1 \\ -2 & -3 & 0 \\ -1 & 0 & 2 \end{pmatrix}$$

```

```
In[29]:= adj12d=Table[( (-1)^(i+j) *S12d[[n12d-i+1,n12d-j+1]] )[[1]],{i,1,4},{j,1,4}];  
MatrixForm[adj12d]
```

Part: Part specification 2[[1]] is longer than depth of object. [i](#)

Part: Part specification 0[[1]] is longer than depth of object. [i](#)

Part: Part specification (-1)[[1]] is longer than depth of object. [i](#)

General: Further output of Part::partd will be suppressed during this calculation. [i](#)

```
Out[30]//MatrixForm=  

$$\begin{pmatrix} 2[[1]] & 0[[1]] & (-1)[[1]] & -1 \\ 0[[1]] & (-3)[[1]] & 2[[1]] & \text{List}[[1]] \\ (-1)[[1]] & 2[[1]] & (-1)[[1]] & -1 \\ -1 & \{-1, -2, -1\}, \{-2, -3, 0\}, \{-1, 0, 2\} & -1 & \text{Symbol}[[1]] \end{pmatrix}$$

```

```
In[31]:= inv12d=Transpose[adj12d]  
Det[m12d];  
MatrixForm[inv12d]
```

```
Out[32]//MatrixForm=  

$$\begin{pmatrix} -2[[1]] & -0[[1]] & -(-1)[[1]] & 1 \\ -0[[1]] & -(-3)[[1]] & -2[[1]] & \{1, 2, 1\}, \{2, 3, 0\}, \{1, 0, -2\} \\ -(-1)[[1]] & -2[[1]] & -(-1)[[1]] & 1 \\ 1 & -\text{List}[[1]] & 1 & -\text{Symbol}[[1]] \end{pmatrix}$$

```